INCH-POUND

MIL-T-8275C
30 MARCH 1994
SUPERSEDING
MIL-T-8275B
7 December 1970

MILITARY SPECIFICATION

# TRANSMITTER, RATE OF FLOW, FUEL SYNCHRO STYLE, TYPE MA-1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1 SCOPE

1.1 Scope. This specification covers one type of remote indicating synchro style fuel flow rate transmitter, designed type MA-1.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

Federal

| | |
|---|---|
| PPP-B-1672 | Boxes, Shipping, Reusable With Cushioning |

Military

| | |
|---|---|
| MIL-P-116 | Preservation, Method Of |
| MIL-B-121 | Barrier Material, Greaseproofed, Waterproofed, Flexible |
| MIL-T-5350 | Transmitters, Synchro, Aircraft, General Specification For |
| MIL-T-5624 | Turbine Fuel, Aviation, grades JP-4 and JP-5 |
| MIL-I-8276 | Indicator Rate of Flow, Fuel Synchro Style, Type MA-2 |
| MIL-B-27497 | Bearing, Jewel, Sapphire Or Ruby Synthetic |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/TICLA, Tinker AFB, OK 73145-5990 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6620

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STANDARDS

Federal

| | |
|---|---|
| FED-STD-101 | Preservation, Packaging And Packing Materials: Test Procedures |

Military

| | |
|---|---|
| MIL-STD-129 | Marking For Shipment And Storage |
| MIL-STD-810C | Environmental Test Methods |
| MIL-STD-2703-1 | DoD Material Procedures For Development And Application Of Packaging Requirements |
| MS33649 | Bosses, Fluid Connection, Internal Straight Thread |
| MS336678 | Connector, Receptacle, Electric, Integral Mounting |

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Ave, Building #4, Section D, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document (s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

American Society For Testing and Material (ASTM)
ASTM D910 Gasolines, Aviation 23 Mar 83

(Application for copies should be addressed to: ASTM, 1916 Race St, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards) the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2) a sample shall be subjected to first article inspection (see 6.2) in accordance with 4.2.

3.2 General specification. The requirements of MIL-T-5350 apply as requirements of this specification with the exceptions and additions specified herein. When the two specifications conflict, this specification shall govern.

3.2.1 Exceptions to the requirements of MIL-T-5350. The altitude, seasoning, and acceleration tests of MIL-T-5350 shall not be required by this specification.

3.3 Design and construction.



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3.3.1 Fuel variation. Unless otherwise specified in the contract or order, the fuel flow transmitter shall be calibrated for rate of flow in pounds per hour (pph) of aviation fuel in accordance with MIL-T-5624. The specific gravity of the fuel shall be 0.770 ± .005 at 15.6°C.

3.3.1.1 Compatibility. The calibration of the transmitter shall be such that it will satisfactorily operate with a type MA-2 indicator in accordance with MIL-I-8276.

3.3.2 Damping. The transmitter shall be provided with suitable damping.

3.3.3 Safety wiring. The transmitter body screws shall be safetied with stainless steel lock wire. Not more than three screws shall be saftied with one length of wire. The lock wire shall be twisted tight between each pair of screws. Suitable lockwashers in place of lockwire may be used, where necessary.

3.3.4 Mounting. Mounting bosses and spacing thereof shall be as shown on figure 1.

3.3.5 Dimensions. The outline dimensions of the transmitter shall conform to figure 1. Connection bosses shall conform to MS33649-20. Electrical receptacles shall conform to MS33678-14s.

3.3.6 Flow direction. The direction of fuel flow through the transmitter shall be permanently and legibly marked by a raised arrow as shown in figure 1, on both top and bottom.

3.4 Performance. The transmitter shall be capable of meeting the requirements of MIL-T-5350 with the following changes:

a. Temperatures - operating temperatures ranging from -55° to +121°C and storage temperatures ranging from -65° to +190°C.
b. Vibration - vibration from .060 double amplitude to ±10g at frequencies form 5 to 500 Hz.
c. Altitude - no requirement.
d. Seasoning - no requirement.
e. Endurance - Cycling between 0 and 80 per cent of full scale for 100,000 cycles at a rate of 15 cycles per minute (cpm).
f. Shock - shock forces up to 15 g's.
g. Acceleration - no requirement.

3.5 Weight. The weight of the transmitter shall not exceed 7.5 pounds.

3.6 Rust inhibitor. A rust inhibitor of a type and quality acceptable to the procuring activity shall be placed in the metering chamber of each transmitter prior to sealing for shipment.

3.7 Caps. The fuel line connections shall be furnished with caps or plugs to prevent foreign material from entering the transmitter.

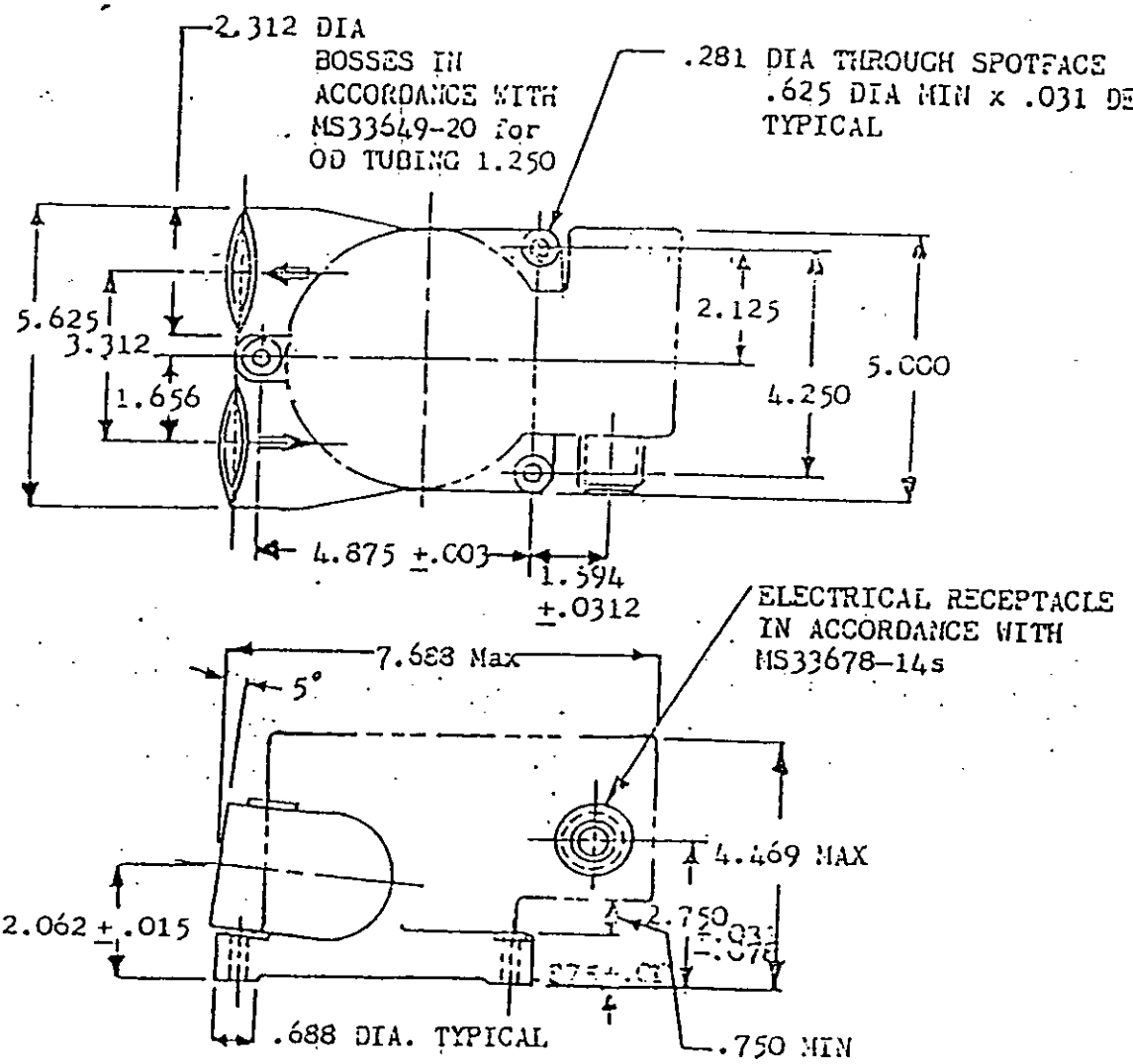
3.8 Synthetic jewel bearings. When synthetic sapphire or ruby jewel bearings are used, they shall be in accordance with MIL-B-27497.

3.9 Recovered materials. Recovered materials shall be used to the maximum extent possible without jeopardizing the end use of the item.

3.10 Cadmium. Cadmium is not to be used in the transmitter unless approved by the preparing activity. There shall be no cadmium used on the outside case of the transmitter. The fluid whose pressure is being measured by the transmitter shall not come into contact with any cadmium in the transmitter.



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DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED,
TOLERANCES: Decimals ±0.005
Angles ±1°

FIGURE 1. Flowmeter transmitter.



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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to insure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 General specification. Sampling, inspection, and test procedures shall be in accordance with MIL-T-5350, except as specified herein.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection. (See 4.5)

b. Quality conformance inspection. (See 4.6)

4.4 Test conditions.

4.4.1 Test position. The transmitter shall be tested with the plane of mounting vertical and with the outlet boss directly above the inlet boss.

4.4.2 Test fuel. Unless otherwise specified, the fuel used for calibration in testing shall be JP-4 conforming to MIL-T-5624. The specific gravity of the fuel shall be 0.770 ± .005 at 15.6°C. (Note: An approved commercial equivalent with higher flash point may be used.)

4.5 First article inspections. First article inspections shall consist of all the tests Section 4 of this specification. Three samples will be tested.

4.6 Quality conformance inspection. The quality conformance inspection shall consist of individual inspection and sampling plan inspections.

4.6.1 Individual inspection. In addition to the individual tests of MIL-T-5350, each transmitter shall be subjected to the leak test (4.7.3) of this specification.

4.6.2 Sampling inspection.

4.6.2.1 Sampling plan A.. The test specified in 4.7.5 shall replace the high temperature test of MIL-T-5350, and the test specified in 4.7.6 shall replace the vibration test of MIL-T-5350.


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4.6.2.2 Sampling plan B. The altitude portion of extreme temperature exposure, seasoning, damping and acceleration tests of MIL-T-5350 shall be omitted. The extreme temperature exposure, endurance and shock tests shall be conducted as specified under 4.7.7, 4.7.8 and 4.7.13 respectively. In addition to the remaining tests listed under sampling plan B of MIL-T-5350, the following tests shall be conducted:

a. Pressure drop (4.7.10)
b. Substituted fuel (4.7.11)
c. Vibration (resonant and cycling) (4.7.12)

4.7 Test methods.

4.7.1 scale error and friction. Scale error shall be determined at each flow rate specified in table I and shall not exceed the room temp tolerances specified in table I. The difference between readings on

TABLE I

Scale error fuel flowmeter transmitter

| Fuel flow pph | Tolerance pph | | |
|---|---|---|---|
| | Room temp | High temp | Low temp and substitut |
| 400 | 50 | 80 | 100 |
| 600 | 40 | 60 | 80 |
| 1,200 | 40 | 60 | 80 |
| 1,800 | 40 | 60 | 80 |
| 2,400 | 40 | 60 | 80 |
| 3,000 | 50 | 80 | 100 |
| 3,600 | 80 | 120 | 140 |
| 4,200 | 80 | 120 | 140 |
| 4,800 | 100 | 150 | 160 |
| 5,400 | 100 | 150 | 160 |
| 6,000 | 120 | 180 | 180 |
| 8,000 | 500 | 800 | 800 |
| 10,000 | 500 | 800 | 800 |
| 12,000 | 500 | 800 | 800 |
| 14,000 | 600 | 900 | 900 |
| 16,000 | 600 | 900 | 900 |
| 18,000 | 800 | 1,200 | 1,200 |
| 20,000 | 900 | 1,300 | 1,300 |
| 22,000 | 1,000 | 1,500 | 1,500 |
| 24,000 | 1,200 | 1,800 | 1,800 |

4.7.2 Position error. Position error shall not exceed the tolerances specified in Table II.

4.7.3 Leak. The transmitter shall be tested for leaks by applying a nonshock hydraulic pressure of 1,600 psi to the inlet connection with the outlet connection bled free of air and capped. The 1,600 psi pressure shall be rapidly applied 5 times at room temperature. After the last application of pressure there shall be no indication of leaks for a period of 5 minutes as shown by a pressure gauge readable to 10 psi or by visible leakage of a colored test fluid. The temperature of the transmitter shall then be reduced to -54° ± 2°C by submersion in a liquid already at this temperature for a period of not less than 30 minutes. (The



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transmitter may be placed in an ambient temperature of -54° ± 2°C for a period of not less than 3 hours as an alternate method of cooling.) While at this temperature, an 800 psi test pressure shall be rapidly applied 5 times. After the last application of pressure there shall be no evidence of leaks for a period of 3 minutes as detected by the same means used for the room temperature portion of this test.

TABLE II

Position error and friction loss tolerances

| Test | Tolerance pph | |
|---|---|---|
| | 400 - 6.000 scale | 6,000 - 24,000 scale |
| Position error | 40 | 200 |
| Friction | | |
| Room temperature | 25 | 200 |
| High temperature | 25 | 200 |
| Low temperature | 50 | 400 |
| Vibration: | | |
| Signal oscillation | 50 | 350 |
| Signal variation | 50 | 350 |

4.7.4 Low temperature. The low temperature test of MIL-T-5350 shall apply Tolerances shall be as specified in tables I and II.

4.7.5 High temperature. The transmitter shall be subjected to an ambient temperature of 121° ± 2° for a period of 4 hours with fuel in the instrument, but not flowing. During the last 30 minutes of this period, fuel at 38° ± 2°C shall be pumped through the transmitter at a rate of 1,000 pph. The transmitter shall then be tested for scale error and friction under the same conditions. Scale error and friction shall not exceed the tolerances specified in tables I and II.

4.7.6 Vibration. The transmitter shall be subjected to vibration for a total of 3 hours on a suitably designed vibration stand at a total excursion of 0.060 inch to 0.070 inch. Vibration shall be such that the transmitter motion is sinusoidal in a plane inclined 45 degrees to the plane of the mounting studs. The vibration frequency shall be 30 Hz. Following the vibration period, the transmitter shall be tested for scale error and friction as specified in 4.7.1.

4.7.6.1 Signal oscillation. The transmitter shall be subjected to the vibration specified in 4.7.6. The frequency of vibration shall be varied gradually from 20 to 50 Hz while the fuel flow is slowly varied through the transmitter range. The magnitude of signal oscillation and variation shall not exceed the tolerances specified in table II. This test may be combined with the test specified in 4.7.6.

4.7.7 Extreme temperature exposure.

4.7.7.1 Low-temperature exposure. The transmitter shall be subjected to the low-temperature exposure test of MIL-T-5350, except that the temperature of both the fuel and the transmitter after the 72-hour period shall be brought up to -54° ± 1°C. Scale errors and friction, while at this temperature, shall not exceed the tolerances specified in tables I and II.

4.7.7.2 High-temperature exposure. The transmitter shall be subjected to ambient temperature at 121° ± 2°C for a period of 4 hours with fuel in the instrument, but not flowing. The ambient temperature of the transmitter shall then be raised to 190° ± 2°C in a period of approximately 10 minutes and maintained at this temperature for 10 minutes. The temperature shall then be reduced from 190° ± 2°C to 121° ± 2°C



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in a period of approximately 10 minutes. This complete cycle shall be repeated 6 times. After the temperature cycling and while the transmitter is maintained at an ambient temperature of 121° ± 2°C, the transmitter shall be tested for scale error and friction and shall meet the tolerances of tables I and II.

4.7.8 Endurance. The transmitter shall be subjected to 100,000 applications of flow to produce an indication from 0 to 80 ± 5 percent of the range of the transmitter and back to zero at a rate of 15 ± 10 cpm. The test specified in 4.7.1 shall be conducted not less than 1 hour after the completion of the endurance cycling, and the transmitter shall not exceed the tolerance specified in table I by more than 40 pph in the low range, and 320 pph in the upper range.

4.7.9 Scale error and friction after environmental tests. The test specified in 4.7.1 shall be performed after each environmental test is completed, but before the test specified in 4.7.8 is run. The tolerances specified in tables I and II shall apply.

4.7.10 Pressure drop. The pressure drop within the transmitter for rates of flow specified shall be tested and shall not exceed the tolerances specified in table III.

TABLE III

Pressure drop tolerances

| Flow rate pph | Maximum allowable pressure drop psi | |
|---|---|---|
| | Normal | Blocked vane |
| 12,000 | 3.5 | 8 |
| 16,000 | 6.0 | 10 |
| 24,000 | 11.0 | 18 |

4.7.10.1 Bypass valve. The pressure drop, with the transmitter movable element blocked in the zero flow position, shall be measured by a suitable method and shall not exceed the values specified in table III.

4.7.11 Substitute fuel. The transmitter shall be subjected to the test specified in 4.7.1, except that the fuel used shall be in accordance with ASTIM D910, and the specific gravity shall be 0.720 ± 0.005 at room temperature. Scale errors shall not exceed the tolerances specified in table I.

4.7.12 Vibration (resonant and cycling). The transmitter shall be subjected to a vibration test in accordance with MIL-STD-810C, method 514, procedure I, curve D. Function of the transmitter shall be checked while scanning for resonant frequencies. Unless otherwise specified by the Government Inspector, fuel flow at a constant rate of 2,400 pph shall be maintained through the transmitter during vibration. The magnitude of signal oscillation and vibration shall not exceed the tolerances specified in table II. At the completion of this test the transmitter shall be subjected to the tests in 4.7.1 and 4.7.3.

4.7.13 Shock. The transmitter shall be subjected to a shock test in accordance with MIL-STD-810C, method 516, procedure I, except as specified herein. The transmitter shall be subjected to three impact shocks of 15 g's amplitude and 7 ± 2 milliseconds duration. One shock shall be applied along each of the three mutually perpendicular axis of the transmitter.

4.8 Inspection and test. Tests of methods of preservation and packaging shall be accomplished in accordance with Section 4 of MIL-P-116 to insure compliance with Section 5 of this specification.



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5. PACKAGING

5.1 Preservation. Preservation shall be level "A" or "C", as specified. (see 6.2)

5.1.1. Preservation-packaging. Unless otherwise specified by the contracting activity, item shall be packaged in Quantity Unit Pack (QUP) of one each. Each item will be provided a preservation method IC-1 In Accordance With (IAW) MIL-P-116.

5.1.2 Cleaning. Item shall be cleaned IAW MIL-P-116. C-1 process.

5.1.3 Drying. Immediately after cleaning, the item shall be dried following any one or combination of the drying procedures listed in MIL-P-116. The drying procedures employed shall not be injurious to the item.

5.1.4 Level A. Item shall be preserved IAW MIL-P-116 and MIL-STD-2073-1 to provide a method IC-1.

5.1.5 Level C Item shall be individually preserved in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity as specified in MIL-P-116 and MIL-STD-2073-1.

5.2 Packing. Packing shall be level "A", "B", or "C", as specified. (see 6.2)

5.2.1 Container-cushioning. Unit container requirements shall conform to MIL-B-121, barrier mateial, unless otherwise specified by the contracting activity. Unit containers shall be large enough to allow for the application of sufficient cushioning/wrap material, between the item and unit container. Item cushioning/wrap material shall be of sufficient density and thickness, to protect the unit container from punctures. Intermediate/shipping containers shall be large enough to allow for application of sufficient cushioning material, between the unit container and intermediate/shipping container. Cushioning material shall be of sufficient density and thickness to insure shock transmission does not exceed peak values in G's established for the item.

5.2.2 Level A. Fiberboard containers do not meet Level "A" container criteria.

5.2.3 Level B. Item will be preserved as specified in 5.1 and shall be packed in exterior containers conforming to PPP-B-1672, weather-resistant unless otherwise specified by the contracting activity. Exterior container shall be uniform shape, size and minimum tare and cube, consistent with the protection required. Closure shall be in accordance with appropriate PPP-B-1672 procedures, as specified by Special Packaging Instructions (SPI), and contracting activity.

5.2.4 Level C. Item will be packed in such a manner that will afford adequate protection against physical damage during direct domestic shipment from the supply source to the first receiving activity.. These packs shall conform to MIL-STD-2073-1.

5.3 Marking. Unit, intermediate, and exterior containers shall be marked IAW MIL-STD-129 and special markings required by the contractor activity.

5.4 Inspection and test. Test of methods of preservation shall be accomplished in accordance with section 4 of MIL-P-116 to insure compliance with section 5 of this specification. Packaging tests shall be conducted IAW rough handling as specified in FED-STD-101.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)



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6.1 Intended use. The Type MA-1 fuel flowmeter transmitter covered by this specification is intended to indicate the rate of flow of fuel in pph being consumed by an aircraft jet engine in conjunction with a suitable indicator.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of the specification.

b. Selection of applicable level of packaging and packing required (See 5.1 and 5.2).

c. Issue of DODISS to be cited in the solicitation, and if required the specific issue of individual documents referenced (See 2.1).

6.3 First article. The specification requires First Article inspections. The organization responsible for the tests is Oklahoma City ALC/TICLA, Tinker AFB., OK 73145-5990.

6.4 Subject term (keyword) listing.

Rate of Flow
Transmitter
Synchro Style

6.5 Change from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Air Force -99
Army -AV
Navy -AS

Preparing Activity:

Air Force -71

Project Number: 6620-0515



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

**INSTRUCTIONS**

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER | 2. DOCUMENT DATE (YYMMDD) |
|---|---|---|
| | MIL-T - 8275C | 94/03/30 |

3. DOCUMENT TITLE
TRANSMITTER, RATE OF FLOW, FUEL SYNCHRO STYLE. TYPE MA-1

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME (Last, First, Middle Initial) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON (If applicable) | 7. DATE SUBMITTED (YYMMDD) |

8. PREPARING ACTIVITY

| a. NAME | b. TELEPHONE (Include Area Code) | |
|---|---|---|
| OC-ALC/TICLA<br>TINKER AFB, OK 73145-5990 | (1) Commercial | (2) AUTOVON<br>336-5960 |
| c. ADDRESS (Include Zip Code) | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 | |



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